

Quantifying Carbon, Water, Cost, and Capability in Peer-to-Peer Inference

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Abstract—Data-centre electricity demand is projected to more than double to approximately 945 TWh by 2030, about the consumption of Japan, with AI a major driver of the growth. A widely proposed remedy is to decentralize it: run small language models (SLMs) on already-owned consumer hardware, connect peers directly over WebRTC, and call frontier cloud models only when a task truly needs them. We ask when this helps and when it does not. We build a parameter-driven model of per-query energy, water, carbon and cost, grounded in public measurements, published estimates and explicit sensitivity ranges, with a fleet-scale extrapolation on the IEA 2030 base and a 50,000-draw Monte-Carlo with one-at-a-time sensitivity analysis; six hypotheses and three counter-claims each receive an explicitly scoped supported, partial, or refuted verdict.

The central result is scoped. For the capability-sufficient fraction of a representative assistant workload - approximately 82% at a 70% acceptance threshold, above the 40–70% agentic-replaceability estimates in NVIDIA’s case studies; our workload mix is an authorial assumption rather than a sampled corpus - edge execution cuts per-query carbon by 54–64% against typical and heavy frontier usage with positive savings in all model draws, nearly eliminates the modeled data-centre on-site cooling-water term, and costs approximately 150 times less per query. The unqualified claim fails against a best-in-class efficient cloud model: compared with Gemini’s 0.24 Wh/prompt a single-user consumer GPU is on average 16% worse, with positive savings in only 3% of draws. The win therefore comes from right-sizing, from eliminating evaporative cooling and PUE/idle overhead (42% of a hyperscaler’s measured per-prompt footprint), and from using marginal energy on existing hardware - not from relocating the same model. The benefit is grid-conditional, survives induced usage growth up to approximately 212% - equivalently, take-back of the entire gross saving, the two being one model in two units - and WebRTC DTLS preserves transport confidentiality against STUN/TURN relays by construction.

Two further passes test this model where it is weakest. A companion study [1] relaxes its central simplification, that per-class capability is fixed by parameter count, and finds f_s unchanged at an ordinary acceptance threshold. A primary measurement campaign across two consumer hardware platforms - an AMD Radeon 8060S APU (64 GB unified memory) and an NVIDIA GeForce RTX 5070 discrete GPU (12 GB GDDR7) - grounds edge decode and prefill in direct observation, with board-energy observation on the NVIDIA platform. On the APU, single-stream generation reaches 78.4, 37.8, and 13.8 tok/s for 3.8B, 8.9B, and 27.3B models, with prompt regressions revealing prefill ratios of 0.050–0.078. A nine-build bandwidth ladder supports the memory-bound decode premise only in its ordering: builds that stream every weight sit below builds that provably do not, by 30 percentage points, but the whole-weight class spans 77–106% of

the specified ceiling rather than a tight band. One dense build exceeds a ceiling no whole-file reader can exceed, which measures the error in the τ -W proxy rather than the hardware and bounds how tightly the statistic can be read. On the discrete GPU, sampled board telemetry measures request-window energy at 0.091 Wh (3B) and 0.19 Wh (8B) at 499 prompt and 300 output tokens, repeated at 0.091 Wh and 0.19 Wh; two significant figures is all the instrument supports. Host-power sensitivities are evaluated up to 100 W. The models, datasets, telemetry harnesses, and an empirical routing agent are included in the reproducibility package, and every measured row is reported rather than filtered. The two hardware targets were operated by the authors: no independent operator or physical-target replication is included, and the AMD concurrent matrix is timing-only because no usable power sensor was available.

Index Terms—sustainable AI, edge inference, small language models, agentic AI, WebRTC, peer-to-peer, data-centre water, carbon footprint, total cost of ownership, Jevons paradox.

I. INTRODUCTION

The marginal cost of an AI answer is no longer purely computational; it is environmental and economic. The IEA projects global data-centre electricity demand to **more than double to ~945 TWh by 2030**, with AI-optimised facilities quadrupling and the United States alone accounting for nearly half of the growth [2]. Per query, a median Gemini text prompt is measured at 0.24 Wh, 0.26 mL water, 0.03 gCO₂e [3]; an infrastructure-aware benchmark estimates a short GPT-4o query at 0.42 Wh and a single long reasoning query (o3, DeepSeek-R1) above 30 Wh [4]. Yet data-centre power-usage effectiveness (PUE) is reported at a global average of 1.54 for six years [5]. Reported water-use effectiveness spans from best-in-class hyperscale operation near 0.12 L/kWh to an industry withdrawal-WUE benchmark around 0.84 L/kWh, with high-water facilities above 1 L/kWh [6], [7].

A decentralized alternative is increasingly credible. NVIDIA Research recently argued that *small language models are the future of agentic AI*: SLMs are “sufficiently powerful, inherently more suitable, and necessarily more economical” for most agentic invocations, and *heterogeneous* systems that mix SLMs with occasional frontier calls are the natural design [8]. They report SLMs matching far larger models: Phi-2 (2.7B) reaching 30B-class commonsense and code at $\sim 15\times$ the speed, SmolLM2 (1.7B) matching 14B peers on tool-calling and instruction-following, and xLAM-2-8B *surpassing* GPT-4o and Claude-3.5 on tool calling. Their case studies estimate

that 40–70% of LLM calls in three agentic systems (MetaGPT, Open Operator, Cradle) are SLM-replaceable; we treat this as author-reported evidence, not as an independent population estimate.

This paper adds a *quantitative environmental, economic, and capability model* for decentralizing such inference to peers’ own devices over relay-confidential transport, with explicit boundary conditions. The *reference architecture* we model throughout is assembled entirely from public components: WebRTC SCTP-over-DTLS transport, STUN-first NAT traversal with geo-spread TURN fallback, and a local-first SLM runtime (Ollama/LiteLLM serving Ministral- and Llama-class 3B–8B builds) with an operator-selectable multi-provider cloud path (OpenAI, Anthropic, Google). We describe it not as a system contribution but as the concrete configuration the model’s parameters refer to, so that every term in the analysis has an unambiguous physical referent. Every quantitative input is drawn from a public cited source or an explicit modelling assumption varied in sensitivity analysis; no proprietary component is required to reproduce any result reported here.

Contributions. (1) A parameter-driven model of per-query energy, water, carbon, and cost for cloud-frontier vs. edge-SLM inference, plus P2P transport (§IV). (2) A capability-sufficiency formulation tied to current public benchmarks and read against NVIDIA’s case-study agentic-replaceability estimates, which it exceeds (§V). (3) A *fleet-scale* extrapolation on the IEA 2030 inference base, and a *total-cost-of-ownership* comparison (§V). (4) Six falsifiable hypotheses and three counter-claims, each with an explicit model-scoped verdict backed by a 50k-draw Monte-Carlo and a sensitivity tornado, including an explicit account of *where the thesis fails* (§VI). (5) A reproducible open harness and a runnable agent that executes the derived routing rule live. (6) *Primary measurement* of single-stream inference on two consumer platforms, a unified-memory APU and a discrete GPU (§V-H), which replaces two of the three borrowed terms in the edge-energy model with measured ones, including board energy on the discrete GPU, and puts the bandwidth-bound premise behind the device-class argument to a test it could fail — and partly did.

We avoid claims outside the evidence. The contribution is the boundary conditions under which decentralization helps, and the cases where it does not.

II. BACKGROUND AND RELATED WORK

The footprint of inference. PUE captures facility overhead (reported global 1.54 [5]; hyperscale 1.09–1.15 in the cited production and survey contexts). The EU Energy Efficiency Directive now mandates data-centre reporting and is preparing a rating scheme and possible minimum performance standards; we therefore do not model an EU-wide PUE cap [9]. WUE for on-site cooling varies widely: AWS reports 0.15 L/kWh for 2024, improving to 0.12 in 2025, and cites an industry withdrawal-WUE benchmark of 0.84 L/kWh [6], [7]; off-site water embedded in generation is separate and often larger [10]; we carry it as an explicit scenario parameter (EWIF, Table I) with a wide range rather than a cited point value, because

published factors vary by generation mix and no single figure fits a global average. Reported grid carbon intensity averages 458 gCO₂e/kWh globally, spanning reported EU ~140 to India ~708 [11]. Jegham *et al.* [4] estimate per-query energy across model sizes; Google [3] provide a comprehensive at-scale measurement showing 42% of the per-prompt footprint is *non-compute* (CPU/DRAM 24%, idle/redundancy/overhead 18%). That is a margin excluded from Google’s cloud boundary, not a guaranteed saving on an edge device whose host, idle, and power-supply draw must be counted separately.

Small models, agentic systems. Surveys of SLMs [12] and the Ministral 3B/8B release [13] document 3B–8B models approaching frontier quality on knowledge, coding, and grounded tasks while trailing on hard reasoning. NVIDIA’s agentic-SLM position paper [8] crystallises this into a design thesis: right-size each invocation and fall back to a frontier model only when needed. It also supplies the capability and economic evidence we build on. Our work is complementary: where [8] argues capability and cost, we add the *environmental* model, the *transport* layer, and a falsifiable boundary analysis.

The current frontier and price sensitivity. Commercial model rosters and prices changed between the measurement corpus and this analysis. We use dated provider pricing pages only as economic inputs; they are not evidence for release chronology, model architecture, or per-query energy [14]–[16]. The energy, water, and carbon results therefore remain pinned to the measured models. In the source set used for this release, we found no public per-query energy measurement for the currently priced models; this is a search result, not proof that no such measurement exists. The decision rule consequently treats the displaced cloud baseline as a parameter rather than silently transferring an older model’s energy to a newer one. That keeps the efficient-cloud counter-case of H5 visible.

Edge inference and P2P transport. Edge execution may cut energy for suitable tasks [17] — a trade-press directional claim that we do not use as a quantitative input — but consumer accelerators can be less efficient *per token* than batched data-centre GPUs, because batching amortises one pass over the weights across concurrent requests and single-stream execution cannot. We make that tension explicit and quantify its mechanism, not its ratio, in C2. We use the STUN/TURN split in Table I only as a scenario informed by a vendor note, not as a measured deployment fraction [18]; the transport is DTLS-secured by mandate with SDP fingerprint pinning [19], [20], so relays forward only ciphertext. This is a transport-layer statement: an authenticated peer is still an application-level endpoint, not a privacy-preserving black box.

Where the measurements are, and are not. Three bodies of work quantify cloud inference energy, and the gap between them is what this paper’s Pass 4 addresses. Jegham *et al.* [4] estimate per-query energy, water and carbon across model sizes using public API data, inferred hardware, and batch assumptions, including a batch-8 scenario on H100/H200-class server accelerators. Elsworth *et al.* [3] report a comprehensive at-scale measurement of a production TPU fleet and are the only source in our set that closes the loop from accelerator

to facility. Caravaca *et al.* [21] contribute over 32,500 direct GPU inference-energy measurements and a predictive per-query model, again on data-centre parts. The first is model-based; the latter two are measurement studies. All three concern the *cloud* side.

The standardised counterpart is MLPerf Inference [22], which defines the single-stream scenario we use and an edge division alongside its datacentre one. Its submissions are vendor-run on curated systems and its edge division has centred on embedded and automotive silicon; we are not aware of results for a consumer unified-memory APU serving quantised LLMs through a commodity runtime, which is the configuration a household actually has.

We are not aware of a comparable measurement of the *consumer* side, and that asymmetry has a consequence for every study of this kind: the cloud baseline is measured while the edge baseline is extrapolated from vendor specifications and enthusiast benchmarks. Ours was, until §V-H. We report empirical measurements across two distinct consumer device classes: a unified-memory APU (AMD Radeon 8060S with 64 GB unified LPDDR5X) and a high-bandwidth discrete GPU (NVIDIA GeForce RTX 5070 with 12 GB GDDR7). These provide execution throughput under commodity runtimes, a multi-architecture bandwidth roofline evaluation, and direct sampled board-power integration. The premise those numbers are used to test is not new in itself. That memory-bound kernels are limited by operational intensity against a bandwidth ceiling is the roofline model [23], and that single-stream autoregressive decode sits on the memory-bound side of that roof is widely assumed in systems work on LLM serving. What is usually missing is a measurement of *where under the roof* a consumer part actually lands when running a commodity stack, across model families it was not tuned for. Reporting that - and reporting the architectures that must and do sit above the roof - is the contribution here. The scope is modest: one serving stack, two machines.

The rebound risk. Efficiency can induce demand (Jevons' paradox); LLM rebound is estimated at 15–25% [24]. We model embodied hardware carbon as a 15–30% sensitivity range, informed by life-cycle accounting work but not directly established by the retrieved source [25]. We charge both against our savings.

III. SYSTEM MODEL AND REFERENCE ARCHITECTURE

Figure 1 shows the decentralized data path. A query is classified; a capability router sends it to a local SLM when the task is SLM-sufficient, an edge device is available, and the local grid is below the routing crossover; otherwise it routes to a cloud frontier provider. STUN establishes direct P2P for the majority of sessions; a geographically co-located (e.g. ISP-edge) TURN relay forwards *ciphertext only* for the NAT-restricted minority. This is the heterogeneous design advocated in [8], with the transport and siting made concrete.

Realization. The architecture is realized across two implemented components, and the analysis distinguishes them throughout. Transport and local-first execution are provided by

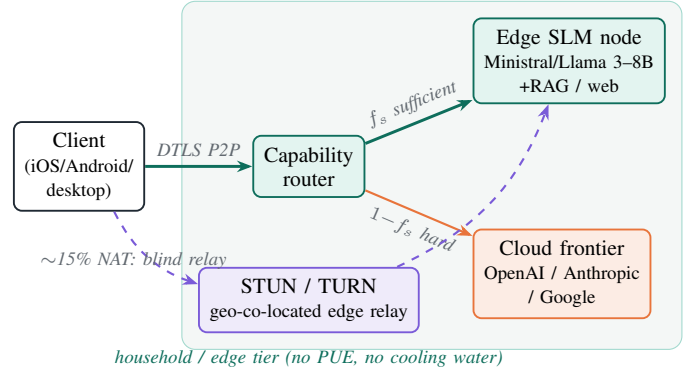


Fig. 1. Decentralized reference architecture. Capability-sufficient queries (f_s) execute on an already-owned edge node and never reach a data centre; the rest fall back to cloud frontier models. P2P DTLS keeps STUN/TURN relays as blind ciphertext forwarders.

the platform server: datachannels run SCTP-over-DTLS, NAT class is determined by STUN probing, and operator-configured ICE servers are merged with short-lived TURN credentials issued by a geographically aware credential service, degrading to local-only configuration whenever that service is unreachable. Its default inference provider is the on-device SLM runtime; cloud providers are configured alternatives, reached either through a direct Gemini adapter or through a LiteLLM adapter for OpenAI- and Anthropic-class models. Per-query capability routing is provided by the reference agent. The division is deliberate: the server treats a locally configured provider as binding and performs no automatic escalation, so that no query leaves the device without an explicit operator decision. The analytical consequence is direct. Results that depend on routing the hard ($1 - f_s$) fraction to a frontier model characterize the heterogeneous system as specified and as implemented in the agent, whereas a default server deployment realizes the f_s edge fraction and delegates escalation to operator policy.

IV. METHODOLOGY

All parameters and credible ranges are in Table I; each carries a citation and is varied in the Monte-Carlo. Energy is per query in Wh, water in mL, carbon in gCO₂e, cost in USD.

A. Per-query footprints

We treat published per-query *cloud* energy as comprehensive at-the-wall energy (it already folds in PUE/idle [3], [4]); we do not double-count PUE. On-site cooling water is charged on IT energy (E/PUE), not total, to avoid overstating data-centre water:

$$W_{\text{cloud}} = \frac{E_{\text{cloud}}}{\text{PUE}} \text{WUE} + E_{\text{cloud}} \text{EWIF}, \quad C_{\text{cloud}} = \frac{E_{\text{cloud}} \text{CI}}{1000}. \quad (1)$$

Edge inference energy for n_o output tokens at throughput τ , board power P , idle baseline P_0 , prefill factor β :

$$E_{\text{edge}} = \frac{P_* (n_o/\tau)(1 + \beta)}{3600}, \quad P_* = \begin{cases} P & \text{full} \\ P - P_0 & \text{marginal} \end{cases} \quad (2)$$

The edge has negligible cooling water ($WUE_{\text{edge}} \approx 0$) but carries grid water and grid carbon, plus amortized embodied carbon $C_{\text{emb}} = 1000 m a / N$ for device mass-carbon m (kg), attributable fraction a , lifetime queries N :

$$W_{\text{edge}} = E_{\text{edge}} \text{PUE}_e \text{EWIF}, \quad C_{\text{edge}} = \frac{E_{\text{edge}} \text{PUE}_e \text{CI}_{\text{loc}}}{1000} + C_{\text{emb}}. \quad (3)$$

B. Right-sizing, capability, and system savings

The right-sizing ratio is $R = E_{\text{frontier}} / E_{\text{SLM}}$. For task class k with capability ratio ρ_k (SLM $\div$ frontier accuracy) and weight w_k , the SLM is *sufficient* iff $\rho_k \geq \alpha$; the sufficient fraction is $f_s = \sum_k w_k \mathbf{1}[\rho_k \geq \alpha]$. The decentralized footprint routes f_s to the edge and the rest to cloud, and charges the rebound r :

$$F_{\text{dec}} = f_s F_{\text{edge}} + (1 - f_s) F_{\text{cloud}}, \quad (4)$$

$$F_{\text{dec}}^{\text{eff}} = F_{\text{dec}} + r (F_{\text{cloud}} - F_{\text{dec}})_+, \quad S = \frac{F_{\text{cloud}} - F_{\text{dec}}^{\text{eff}}}{F_{\text{cloud}}}. \quad (5)$$

C. Cost and fleet-scale

Per-query cost is electricity plus amortized hardware at the edge versus token pricing in the cloud:

$$\$_{\text{edge}} = \frac{E_{\text{edge}}}{1000} p_e + \frac{c_{\text{dev}} a}{N}, \quad \$_{\text{cloud}} = \frac{n_o p_o + n_i p_i}{10^6}. \quad (6)$$

Fleet-scale annual savings scale the IEA 2030 inference base $E_{\text{inf}} = D \cdot s_{\text{AI}} \cdot s_{\text{inf}}$ by adoption η and the realised system energy-savings fraction S_E . Note that S_E already carries f_s through (5), so the sufficient fraction is not applied a second time here:

$$\Delta E = E_{\text{inf}} \eta S_E, \quad \Delta C = \Delta E \cdot \text{CI}. \quad (7)$$

D. Network and Monte-Carlo

For session payload b at intensity ε , a centralized path costs $2\varepsilon b$; the P2P expectation mixes a direct ($1\times$) path for p_d and relayed ($2(1+\theta)\times$) paths otherwise. We draw 50,000 samples with independent triangular distributions over each analyst-declared parameter range and report mean, percentiles, and $P(S > 0)$ per cloud baseline, plus a one-at-a-time tornado. Here $P(S > 0)$ is the fraction of declared model draws with positive savings, not a confidence level or posterior probability about the real world.

V. RESULTS

Table II summarizes all verdicts; we expand the key ones.

A. H2: Right-sizing dominates (Supported within model)

Per-query inference energy spans nearly three orders of magnitude (Fig. 2). An 8B SLM uses $2.97\times$ less energy than GPT-4o long-context and $65\times$ less than o3. The single largest sustainability lever is therefore not *where* a model runs but *which size* runs.

TABLE I
KEY MODEL PARAMETERS (CENTRAL [LOW, HIGH]) AND SOURCES.

Parameter	Value [range]	Source
PUE (hyperscale / global)	1.10 / 1.54	reported contexts; [3], [5]
WUE on-site (hyperscale)	0.20 [0.12, 0.30] L/kWh	model; informed by [6]
EWIF (grid water)	1.10 [0.50, 2.50] L/kWh	assumption
Grid CI (global)	458 [400, 473] g/kWh	reported input; [11]
E Gemini / GPT-4o / o3	0.24 / 1.79 / 39.2 Wh	[3], [4]
E Llama-3.1 8B (cloud)	0.60 Wh	[4]
RTX 4090 (8B) throughput	141 tok/s	legacy scenario; [26]
RTX 4090 board power	420 [350, 450] W	assumed; 450 W cap [27]
Radeon 8060S (8.9B) tput	37.8 tok/s	measured The
Prefill overhead β	0.055 [0.050, 0.078]	measured
Desktop idle power	60 [30,150] W	sensitivity assumption; [28]
STUN direct / TURN relay	0.80 / 0.15	scenario; [18]
Net intensity	0.015 [0.006,0.06] kWh/GB	sensitivity; informed by [29]
Rebound factor	0.20 [0.15,0.25]	[24]
Cloud out/in price	\$15 / \$3 per Mtok	dated price snapshots; [14]–[16]
Electricity price	\$0.16 [0.10,0.30]/kWh	est.
2030 DC demand / AI share	945 TWh / 0.45	IEA; share is an assumption [2]
α sufficiency	0.70 [0.60,0.80]	this work

Gemini value is Google’s measured median; the GPT-4o, o3, and Llama-3.1 values are Table 4 values from Jegham v2’s model-based estimate, not measurements from this study.

B. H3: “ $\sim 70\%$ capable” is true only when scoped (Partial)

At $\alpha = 0.70$ the SLM is sufficient for extraction/routing ($\rho=0.95$), RAG-grounded QA (0.90), summarization (0.88), and simple code (0.75), but *not* hard reasoning (0.51). The weighted sufficient fraction is $f_s = 82\%$ (Fig. 3). The *blanket* claim is **refuted**; on hard math the 8B/frontier ratio is $\approx 0.51 < 0.70$.

Where these five numbers come from, stated plainly. They are *authorial estimates*. They are not read off a table in any work we cite. What informs them is a literature we can point at but not compute from: public benchmarks put Ministral-8B at MMLU 65.0, HumanEval 76.8 and GSM8K 54.5 against Llama-3.1-8B [13] — vendor-run, on the vendor’s own evaluation protocol, which differs from the protocol behind commonly quoted Llama-3.1-8B figures and should not be read as a head-to-head; surveys of small models report 3–8B builds approaching frontier quality on grounded and structured work and falling away on multi-step reasoning [12]; xLAM-

TABLE II

HYPOTHESIS AND COUNTER-CLAIM VERDICTS. SYMBOLS CARRY THE VERDICT INDEPENDENTLY OF COLOUR: ✓ UPHOLDS THE THESIS, ~ UPHOLDS IT ONLY UNDER STATED CONDITIONS, × REFUTES THE STATED CLAIM, AND ▲ MARKS A COUNTER-CLAIM THAT SURVIVES.

ID	Statement (abbrev.)	Verdict
H1	P2P lowers network transmission energy	✓ Supported (model)
H2	Right-sizing cuts inference energy 3–65×	✓ Supported (bounded)
H3	SLM $\geq 70\%$ capable on a task class	~ Partial
H4	Edge offsets water + overhead per query	~ Partial
H5	Integrated system nets positive savings	~ Partial
H6	Relay plaintext access prevented (DTLS)	✓ Supported (protocol)
C1	Rebound cancels savings	~ Modelled (conditional)
C2	Consumer GPU less efficient/token	▲ Supported (mech.)
C3	Embodied carbon dominates	▲ Supported (cond.)

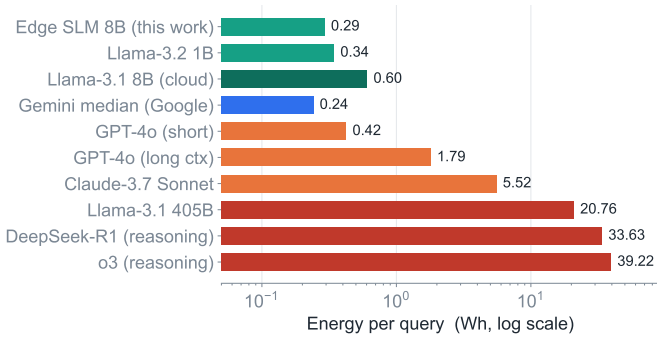


Fig. 2. Per-query inference energy (log scale). A sufficient SLM avoids the 3–65× energy of frontier/reasoning models [3], [4].

2-8B surpasses GPT-4o on tool calling [8]. Only ρ_{hard} has an arithmetic basis: $0.40/0.78 = 0.51$ from reported hard-math accuracies, carried with the contamination range that survey work on benchmarks documents [30]. The other four are judgement.

The workload mix has no source at all. The five class weights (extraction 0.18, RAG QA 0.30, summarization 0.22, simple code 0.12, hard reasoning 0.18) are not sampled from any query corpus, ours or anyone else’s. We did not have one. This matters more than any other unsourced quantity in the paper, because once the ratios are fixed f_s is nothing but $1 - w_{\text{hard}}$: $f_s = 82\%$ **is the assertion that 18% of assistant work is hard reasoning**, and every saving reported here is proportional to it.

This is why f_s sits outside the only external check available. NVIDIA’s case studies estimate 40–70% of calls in three real agentic systems as SLM-replaceable [8]. Our 82% is *above that range*, not at its edge. The discrepancy has a plain reading: MetaGPT, Open Operator and Cradle are agentic developer systems whose traffic is weighted toward planning and code, whereas our mix asserts a consumer assistant weighted toward retrieval and summarization. That reading may be right, but it is a story told about an assumption, not evidence for it. A reader who prefers NVIDIA’s reported

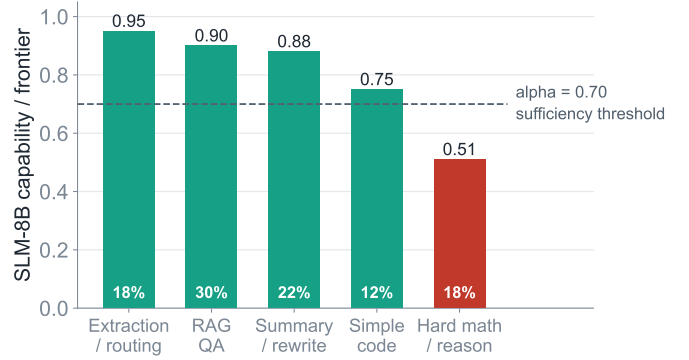


Fig. 3. Capability sufficiency by task class against the $\alpha = 0.70$ threshold. Green routes to the edge; red to cloud. The figure inside each bar is that class’s share of the workload. $f_s = 82\%$.

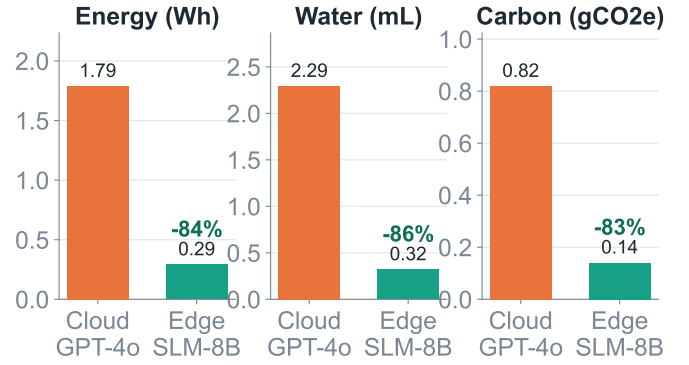


Fig. 4. Per-query footprint for a capability-sufficient task: cloud frontier vs. edge SLM.

range should read their own answer off Fig. 9, where f_s is now swept across $[0.40, 0.90]$ as a first-class factor: at the midpoint of NVIDIA’s reported band ($f_s = 55\%$) carbon savings against the GPT-4o long baseline fall from 54.4% to 36.5%, and at its floor ($f_s = 40\%$) to 26.5%; the fleet figures scale with them. The *sign* of the result survives the whole range against a frontier baseline, and that is the claim we make. Its *magnitude* does not.

C. H4: Edge offset is water-robust and carbon-conditional (Partial)

For a capability-sufficient query, the model-based cloud frontier (GPT-4o) estimate costs 1.79 Wh, 2.29 mL, 0.82 gCO₂e; an edge SLM request costs 0.29 Wh, 0.32 mL, 0.14 gCO₂e (Fig. 4). Water savings are robust because edge cooling is ambient. Carbon savings are *conditional on the local grid*: Fig. 5 shows that an edge node on a coal-heavy grid (India ~ 708 g/kWh) can emit more than a clean cloud region (EU ~ 140) running the same model.

D. H5: System savings validate against frontier use, not efficient cloud (Partial)

With $f_s = 82\%$ and the literature rebound of 20% take-back (equivalently 42% induced usage, (12)), the decentralized

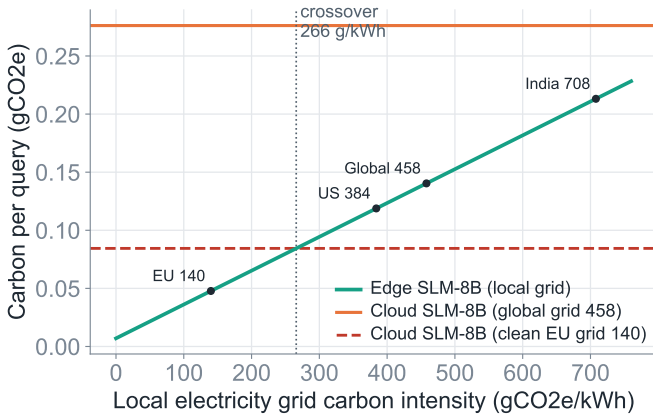


Fig. 5. Edge carbon scales with the local grid; it beats a clean cloud region only below the crossover intensity.

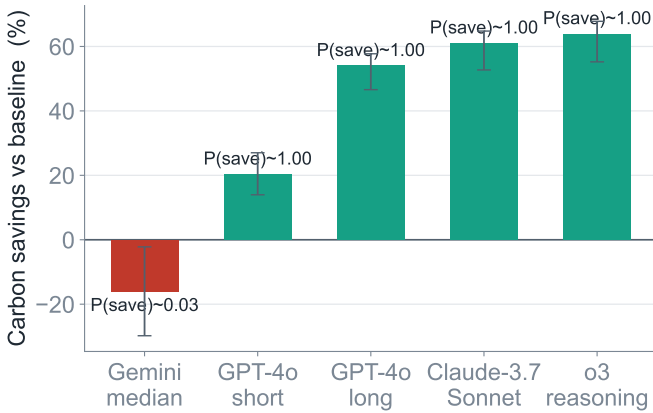


Fig. 6. Carbon savings (mean, 5–95% Monte-Carlo) vs. five cloud baselines. Positive only against frontier usage; refuted against best-in-class efficient cloud (Gemini, red).

system saves 54.9% energy, 56.4% water, and 54.4% carbon per query against a GPT-4o long-context baseline. Those are point estimates at every parameter’s central value; Table III gives the corresponding Monte-Carlo means (+54.0% carbon), which differ slightly because the draws span asymmetric parameter ranges rather than sitting at their midpoints. They are the same quantity under two estimators, labelled as such rather than reported as whichever is larger. Within these sampled model assumptions, the Monte-Carlo (Fig. 6, Table III) returns positive savings against typical and heavy frontier baselines ($P(S > 0) = 1.00$). **The counter-case:** against Gemini’s hyper-optimized 0.24 Wh prompt the system is on *average* 16% worse ($P(S > 0) = 0.03$). The thesis is about displacing *frontier* usage, not about beating an already-efficient small cloud service.

E. Economics: $\sim 150\times$ cheaper per query

Cost reinforces the environmental case (Fig. 7). On an already-owned device an edge query costs electricity only ($\sim \$4 \times 10^{-5}$); a cloud mid/frontier query is $\sim \$6 \times 10^{-3}$, about $150\times$ more, and a reasoning-model query more still.

TABLE III
MONTE-CARLO CARBON SAVINGS VS. CLOUD BASELINE ($n=50K$).

Baseline	Mean	p5	p95	$P(S>0)$
Gemini median	−16.1%	−29.8%	−2.2%	0.026
GPT-4o short	+20.4%	+14.0%	+27.0%	1.000
GPT-4o long	+54.0%	+46.6%	+57.7%	1.000
Claude-3.7	+61.0%	+52.7%	+64.8%	1.000
o3 reasoning	+63.9%	+55.2%	+67.8%	1.000

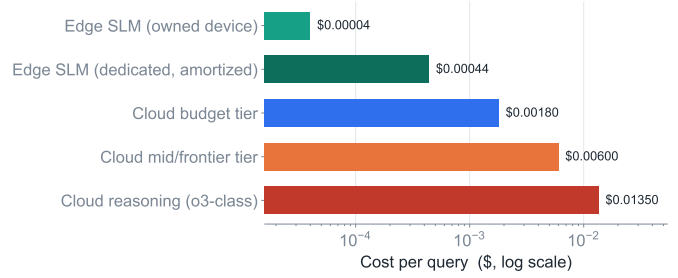


Fig. 7. Per-query cost (log scale): edge vs. cloud tiers.

Even a *dedicated* amortized device ($\sim \$4.5 \times 10^{-4}$) is an order of magnitude cheaper than frontier cloud. This independently corroborates NVIDIA’s $10\text{--}30\times$ SLM economy [8] and is, for many deployers, an important driver.

We re-price this comparison with dated first-party provider price snapshots [14]–[16]. These pages are used only as price inputs, not as evidence about release chronology, model architecture, or energy. Across the low, middle, and premium price inputs, the calculated edge-to-cloud ratio moves materially while preserving the same ordering: decisive against expensive frontier tiers and materially narrower against efficient ones. The exact ratio is therefore a scenario output, not a market-wide statistic.

F. Fleet-scale upside

On the IEA 2030 base (945 TWh data-centre demand, $\sim 45\%$ AI, $\sim 65\%$ inference, i.e. ~ 276 TWh of AI inference), shifting the sufficient fraction to the edge at 25% adoption avoids ~ 38 TWh and ~ 17.4 MtCO₂e per year, equivalent to the annual electricity of ~ 3.5 million U.S. homes; at 50% adoption, ~ 34.8 MtCO₂e (Fig. 8). These are upside bounds contingent on the H5 conditions holding.

G. Sensitivity: what moves the result

A one-at-a-time tornado (Fig. 9) over each parameter’s credible range shows the carbon-savings result (vs. GPT-4o long) is dominated, by a wide margin, by the *workload mix* through f_s — the one input in the figure with no source at all (\$V-B). Across $f_s \in [0.40, 0.90]$ savings run from 27% to 60%. The acceptance threshold α is second (46–54%) and rebound third (51–58%); grid intensity has no material effect on this baseline because it scales cloud and edge together. No single parameter flips the *sign* against a frontier baseline, which is the robust claim. The *magnitude* is not.

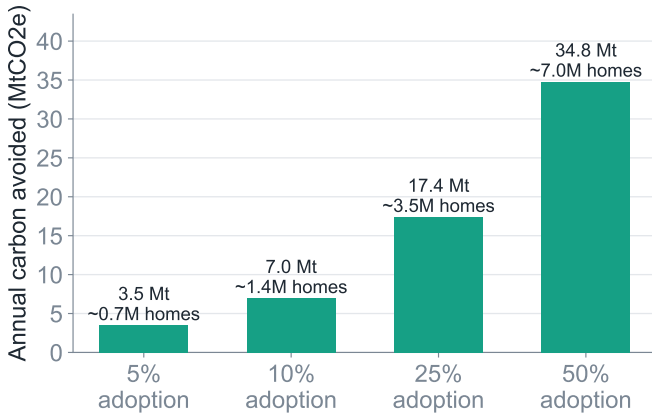


Fig. 8. Annual carbon avoided vs. adoption on the IEA 2030 inference base.

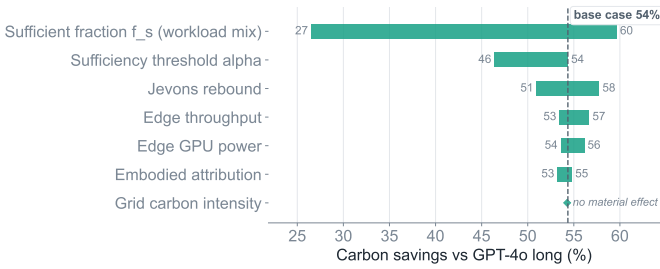


Fig. 9. One-at-a-time sensitivity of carbon savings (vs. GPT-4o long). The workload mix, through f_s , dominates every other parameter and is the only one here with no source; its low end is the floor of NVIDIA’s reported agentic-replaceability range and its centre is this study’s own assertion.

H. Primary measurement on consumer hardware

Every figure above inherits three borrowed terms from the edge-energy model of (2): throughput τ from a third-party benchmark of an RTX 4090, the prefill factor β from our own assumption, and board power P from a vendor TDP. None of the three describes hardware we had touched. We therefore ground these terms in direct empirical measurements across two consumer platforms: an AMD Radeon 8060S integrated APU (Ryzen AI MAX+ 395, 64 GB unified memory) and an NVIDIA GeForce RTX 5070 discrete GPU (12 GB GDDR7).

Execution protocol. On both platforms, inference is executed via Ollama (version 0.33) in single-stream generate mode with raw continuation prompts [31], a fixed pseudo-random seed, temperature zero, and a context window of 4096 tokens. A warm-up generation is discarded before recording. For generation throughput and prefill characterization, prompt length is systematically *swept* across multiple prompt sizes rather than fixed, because prefill duration consists of an amortized fixed per-request overhead a plus a marginal per-token cost b : dividing total prefill duration by prompt length yields $n/(a + bn)$, an apparent rate that varies with n rather than reflecting intrinsic hardware throughput. We regress prefill time on prompt length across four lengths per model and report the marginal rate $1/b$ and fixed overhead a separately ($R^2 \geq 0.9998$ across all fits).

TABLE IV
MEASURED SINGLE-STREAM INFERENCE, AMD RADEON 8060S, 64 GB UNIFIED, Q4_K_M, TEMPLATE BYPASSED. MEDIAN OF FIVE REPETITIONS; PREFILL FROM A FOUR-LENGTH REGRESSION ($R^2 \geq 0.9998$ THROUGHOUT). β IS EVALUATED AT THIS STUDY’S TYPICAL QUERY SHAPE (500 PROMPT, 300 GENERATED TOKENS).

Model	Params	Generation (tok/s)	Prefill		β
			marginal	fixed	
Ministral-3 3B	3.8B	78.4	2,711 tok/s	5.8 ms	0.050
Ministral-3 8B	8.9B	37.8	1,161 tok/s	7.4 ms	0.055
Qwen3.8 27B	27.3B	13.8 [†]	341 tok/s	233 ms	0.078

[†]This build carries a multi-token-prediction head (`nextn_predict_layers=1`), so it emits more than one token per forward pass when its draft is accepted. Its rate is correspondingly unstable across repetitions (12.8–24.5 tok/s, $\sigma = 4.9$) where the two dense models hold to $\sigma/\mu < 0.3\%$. We report the median and the spread rather than suppressing either.

Two caching traps, and why the sweep exposes them.

Prompt caching can silently distort benchmarking by presenting an artificially elevated throughput figure. Repeating an identical prompt allows the serving runtime to reuse the key-value (KV) cache while still reporting the full prompt length: our isolation probe for this mechanism measured an apparent 22,979 tok/s prefill over 640 reported tokens on the 8B model. Injecting a unique cryptographic salt per repetition successfully neutralizes whole-prompt reuse. However, this does not eliminate the second caching trap: the chat template’s system prompt sits ahead of the user text, remains identical across requests, and is therefore cached while still counted in reported token totals. On Ministral-3, 552 of the 639 reported prompt tokens are billed but never recomputed, causing templated execution to measure 7,169 tok/s against 1,106 tok/s when bypassing the template - an artificial inflation of 6.5 $\times$. Fitting against templated endpoints produces a non-physical negative intercept (−482 ms on the 8B), whereas bypassing the template yields a physically consistent +7.4 ms per-request overhead while the true marginal rates agree to within 2.4%. Evaluating models with short system templates (such as Qwen3.8, with only 10 system tokens) provides a natural negative control with no observed cache distortion.

Prefill scaling. We originally assumed $\beta = 0.15$ in (2). Evaluated at the study’s standard 500-in/300-out query shape, empirical prefill factors range from 0.050 to 0.078 on the APU (Table IV), roughly one-third of the assumed value. On the RTX 5070, fitted prefill rates of approximately 9,139 tok/s (3B) and 4,088 tok/s (8B) yield empirical prefill ratios of 0.0404 and 0.0453. Because β scales the fraction of request duration spent on prompt processing, the assumed $\beta = 0.15$ conservatively overstated edge energy consumption. We retain the conservative baseline in our fleet-scale simulations, as long-context retrieval queries will exhibit higher prefill overhead.

The memory bandwidth ceiling across platforms. The core architectural premise of local autoregressive decoding is that single-stream generation is fundamentally bounded by memory bandwidth rather than arithmetic compute [23]. For a

model streaming its entire quantized weight set W per token at decoding throughput τ , the effective read bandwidth is

$$B_{\text{eff}} = \tau \cdot W, \quad u = B_{\text{eff}}/B_{\text{spec}}. \quad (8)$$

If the proxy W were exactly the bytes read per token, this would establish a hard ceiling: no dense architecture reading all its weights could exceed B_{spec} , while architectures reading only a fraction of their weights per token (Mixture-of-Experts) or emitting several tokens per pass would necessarily appear above it. That is the prediction we test. It holds for the ordering and fails for the magnitude, because W is a proxy and not a measurement — which is what the excess above the ceiling measures.

We evaluate this roofline across both platforms (Fig. 10) over *every* build installed on each machine, including builds whose weights are not redistributable. Those rows are marked rather than dropped: that a reader cannot independently check the file size the statistic divides by is a reason to label a row, not a reason to remove it from the evidence. The ladder is nine rows.

On the AMD Radeon 8060S (specified at 256 GB/s via LPDDR5X-8000) [32], nine builds spanning 2.95 to 23.94 GB were benchmarked, five of them dense. Those five span **76.7–105.7%** of the specified bandwidth (median 89%). The three redistributable dense rows do agree closely — 232.2, 227.7 and 226.3 GB/s, within 2.6% across a $6.1\times$ spread in weight size [33], [34] — but that agreement is a property of those three builds, not of the class: a 3.1B Q8_0 build reads 76.7%, and a 7.6B Q8_0 vision-language build reads **105.7%**.

That last row is the useful one, and it refutes the strong form of our own claim. It is dense by the server’s own metadata — no expert routing, no multi-token head — so it streams its whole weight set once per token and *cannot* exceed the memory ceiling. Measuring it above 100% therefore measures the *proxy*, not the hardware: $\tau \cdot W$ counts every byte the file holds, and the file holds a vision tower, embedding and output tables, and quantisation metadata, none of which is read once per token. The excess puts a hard floor of 5.7 percentage points under the proxy’s error, and the error is plainly build-dependent rather than constant — Qwen3 4B reads 63.1% on the RTX 5070, which is what a large vocabulary does to a small weight file, not a compute-bound decode. This statistic therefore does not resolve utilisation to the percentage point: a whole-file reader is measured above a ceiling it cannot physically exceed.

What survives is the *ordering*, and it survives with margin. Every build that provably reads only part of its weights sits above every build that does not: 136% for multi-token prediction, 212% for per-layer embeddings, and 410% and 582% for the two sparse MoE builds (8 of 128 and 8 of 256 experts active). Measured from the highest dense row — the one above the ceiling, not the highest convenient one — the two classes are separated by **30** percentage points, five times the proxy error the same data exposes. Measuring the gap from the highest *redistributable* dense row instead would widen it

to 45 points, which would be selection on the outcome. The decode-class distinction holds; the tight-band reading does not.

On the NVIDIA GeForce RTX 5070, with 12 GB of GDDR7 rated at 672 GB/s [35], Ministral-3 3B reaches 588.6 GB/s (87.6%) and Ministral-3 8B reaches 626.1 GB/s (93.2%), while Qwen3 4B reaches 424.3 GB/s (63.1%). This machine carries no partial-reader control, so it cannot test separation at all, and is reported without a separation claim. Its three dense builds themselves span 30 points of utilisation (63.1–93.2%) — a spread *within* one decode class, not the between-class gap discussed above, and the same proxy limitation restated. Separately from the roofline, the two device classes differ in the way that matters for what a household can run: a discrete GPU offers high bandwidth against a 12 GB capacity wall, while a unified APU trades bandwidth for a 64–128 GB pool that will hold a 27B build at all.

Measured GPU board energy on NVIDIA RTX 5070. While the AMD APU platform provides unified memory capacity, its hardware telemetry does not expose instantaneous power rails without kernel-level privileges. In contrast, the NVIDIA GeForce RTX 5070 exposes continuous board power telemetry via the NVIDIA System Management Interface (`power.draw.instant`) [36]. We request a 0.10-second sampling interval between driver reads and observe roughly 0.16-second spacing.

The driver does not refresh that counter as fast as we poll it, and this bounds the precision of everything below. Consecutive reads return byte-identical values in runs of three to four, so the underlying counter updates on the order of every 0.4–0.5 s. A 3B request returns 13 samples carrying only **4–5 distinct readings** across a 1.6 s window; an 8B request returns 22 samples carrying 6–8. The piecewise-linear interpolation below therefore interpolates a staircase, and the timing of the clock-ramp early in each request is resolved only to about half a second. Combined with a run-to-run spread of 16% on the 3B (7 repetitions, coefficient of variation 6.2%), **two significant figures is the most these measurements support**, and we report them as such throughout.

To avoid edge-truncation errors, the timed inference interval $[0, T]$ is strictly bracketed by telemetry samples immediately preceding and following execution; note that the leading bracket sample can still carry the previous repetition’s load, so the first interpolation interval is the least trustworthy in each trace. Let $\hat{P}(t)$ denote the piecewise-linear interpolation of sampled power. The request-window GPU board energy is computed via numerical integration:

$$E_{\text{GPU}} = \frac{1}{3600} \int_0^T \hat{P}(t) dt. \quad (9)$$

Energy outside $[0, T]$ is not charged, and mean request power is given by $3600E_{\text{GPU}}/T$.

We conduct two experimental passes (a primary run and a same-target repeatability run) comprising seven repetitions per model at a standard query workload of 499 prompt tokens and 300 generated output tokens, ensuring complete GPU residency (Table V and Fig. 11). For Ministral-3 3B, median request

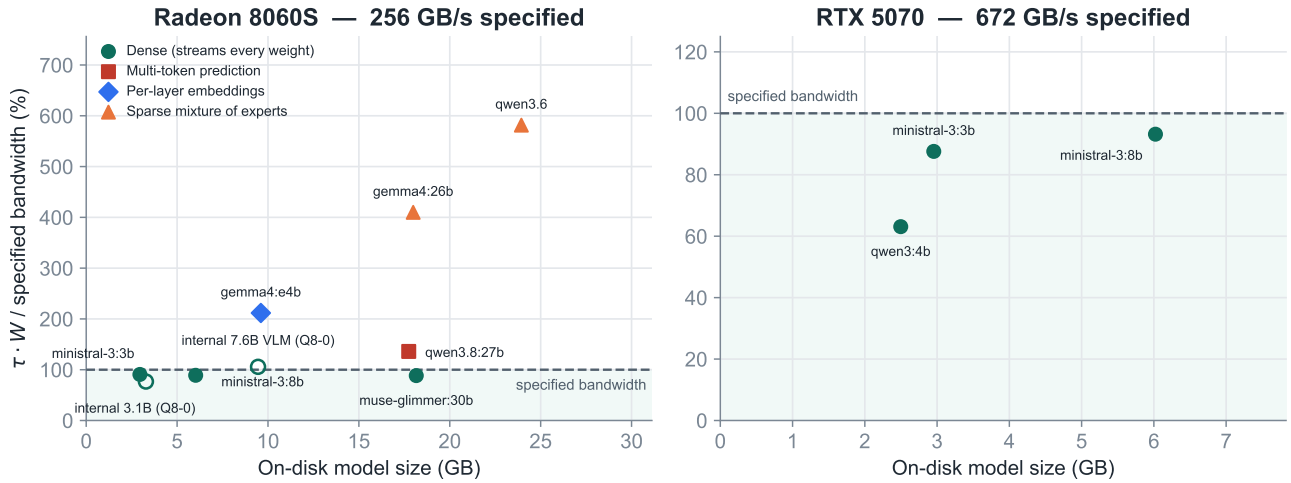


Fig. 10. Effective memory read bandwidth proxy $\tau \cdot W$ normalized to each machine’s own hardware specification: all nine builds measured on the AMD Radeon 8060S (256 GB/s) and three on the NVIDIA GeForce RTX 5070 (672 GB/s). Hollow markers are builds whose weights are not redistributable; they are labelled rather than dropped, because one of them is the row that bounds the proxy. Sparse MoE, multi-token-prediction and per-layer-embedding builds sit above the ceiling because they stream only part of the weight file per token. One *dense* build also sits above it, at 105.7%, which a whole-file reader cannot do: that excess measures the error in $\tau \cdot W$, not the hardware. Marker shape encodes the decode class independently of colour.

TABLE V
MEASURED REQUEST-WINDOW GPU BOARD ENERGY ON NVIDIA
GEFORCE RTX 5070 (499 INPUT / 300 OUTPUT TOKENS, SEVEN
REPETITIONS PER ROW, Q4_K_M).

Model	Run	tok/s	Mean W	Wh	Range (Wh)
Ministral-3 3B	Primary	195.6	203.4	0.091	0.078–0.093
Ministral-3 3B	Repeat	196.4	204.6	0.091	0.078–0.094
Ministral-3 8B	Primary	101.8	224.1	0.19	0.18–0.19
Ministral-3 8B	Repeat	102.5	222.2	0.19	0.17–0.20

board energy is 0.091 Wh (observed range 0.078–0.093 Wh) at 195.6 tok/s, with mean active power of 203.4 W over a 1.62 s window, yielding 0.30 mWh per output token. The repeatability run agrees these results at 0.091 Wh (0.078–0.094 Wh) and 196.4 tok/s. For Ministral-3 8B, median request board energy is 0.19 Wh (0.18–0.19 Wh) at 101.8 tok/s, with mean active power of 224.1 W over 3.11 s, yielding 0.65 mWh per output token. The repeatability run again measures 0.19 Wh, over the range 0.17–0.20 Wh, at 102.5 tok/s.

Host-power sensitivity and accounting boundaries. Crucially, board-level GPU telemetry captures only the active accelerator draw during inference; it does not reflect wall-socket electricity consumption. Total system energy includes host CPU and memory activity P_h and power-supply conversion loss. Writing η for supply efficiency:

$$E_{\text{system}} = \frac{1}{\eta} \left(E_{\text{GPU}} + \frac{P_h \cdot T}{3600} \right). \quad (10)$$

Table VII evaluates this across host power additions of 30 W, 60 W and 100 W at $\eta = 0.90$, a typical 80 PLUS Gold figure at the partial load these requests draw. At a 60 W host addition, 8B request energy rises from 0.19 Wh at the board rail to 0.27 Wh at the wall. Supply conversion loss is charged at η ;

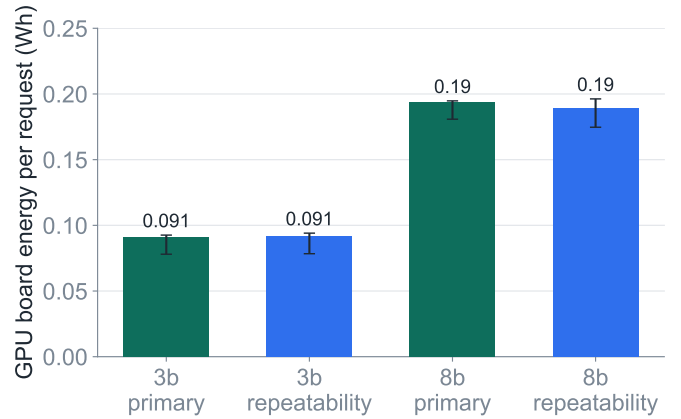


Fig. 11. Measured request-window GPU board energy on NVIDIA GeForce RTX 5070. Bars depict medians across seven repetitions; whiskers depict observed minima and maxima. Host platform draw, PSU losses, and post-request idle tails are evaluated via sensitivity analysis.

the idle tail is not, because post-request standby is a property of a deployment’s duty cycle rather than of a query, and folding an assumed one into a per-query figure would import an assumption under the heading of a measurement. It stays outside the boundary, named here so the boundary is legible. Furthermore, while these measurements establish an empirical local operating budget, token throughput under a synthetic continuation workload does not constitute proof of output quality parity with frontier cloud services. Sustainable AI claims must remain strictly bounded by task sufficiency, system accounting boundaries, and regional grid carbon intensity.

I. A bookkeeping check on the edge-energy model, and what it cannot show

Every headline in §V descends from (2). On a machine that exposes a board rail we can measure P , τ , β and the per-query energy on the same request, and it is worth stating precisely what comparing them does and does not establish.

What it establishes: the terms are consistent. Substituting the measured P , τ and β into (2) and comparing against the separately integrated energy gives a residual of at most 2.3% across six rows — two models, two query shapes, and a same-target repeatability run for each. β is evaluated from the fitted prefill line at each row’s own prompt and output length rather than reused from another shape. All six residuals are negative and small.

What it does not establish: that (2) is right. The comparison above is not a falsification test, and the reason is worth spelling out, because the appearance of a test is worse than no test. Mean request power is as $\bar{P} = 3600 E_{\text{GPU}}/T$, so $E_{\text{GPU}} \equiv \bar{P}T/3600$ identically. Throughput is defined as $\tau = n_o/t_{\text{decode}}$, and β as $t_{\text{prefill}}/t_{\text{decode}}$. Substituting all three into (2) reduces the comparison to

$$t_{\text{prefill}} + t_{\text{decode}} \stackrel{?}{=} T, \quad (11)$$

which is an accounting identity up to the fixed per-request overhead. The residual we report *is* that overhead — 5–8 ms divided by a 1.6–3.1 s window — which is why every one of the six is negative and why none could have been large. The check cannot fail by more than the term it omits, so passing it carries no evidence about whether energy is really power times occupancy.

It is still worth running, for the narrower thing it does prove: that no substantial energy is being spent inside the request window outside prefill and decode, that the harness’s timing and power instruments agree with each other, and that no transcription error separates the two. That is a consistency audit, not a validation, and we now report it as one. The functional form of (2) remains an assumption — a defensible one, but untested by this paper.

The inputs, by contrast, are testable — and were wrong. This is the part of the comparison that carries information, precisely because each term is measured against an independent quantity rather than against its own definition. Table VI compares each assumed term against what the same quantity measures on an 8B request. All three are wrong, and two of them are wrong in the direction that flatters the thesis. The assumed board power exceeds the measured draw by 1.89 $\times$ and the assumed prefill factor exceeds the measured one by 3.38 $\times$; both inflate modelled edge energy. The assumed throughput exceeds the measured one by 1.38 $\times$, which deflates it. The errors partly cancel, which is why the modelled 0.29 Wh per query sits within a few percent of the 0.31 Wh we measure once a 100 W host allowance is charged, while overstating the accelerator-only figure of 0.19 Wh by about half. *A model can agree with reality at the level of its output while being wrong*

TABLE VI
ASSUMED VERSUS MEASURED TERMS OF (2), 8B REQUEST ON THE RTX 5070. “EFFECT” IS THE DIRECTION EACH OVERSTATEMENT PUSHES MODELLED ENERGY.

Term	Assumed	Measured	Ratio	Effect on E
Board power P (W)	420	222.25	1.89 $\times$	raises
Throughput τ (tok/s)	141	102.49	1.38 $\times$	lowers
Prefill factor β	0.15	0.0444	3.38 $\times$	raises

TABLE VII
HOST-POWER SENSITIVITY ANALYSIS ADDED TO PRIMARY GPU BOARD MEASUREMENTS.

Added Host Power	3B Energy (Wh)	8B Energy (Wh)
0 W (Board only)	0.091	0.19
30 W	0.12	0.24
60 W	0.13	0.27
100 W	0.15	0.31

in every term, and only measuring the terms separates the two cases.

What we changed, and what we did not. We leave (2) and its parameters unchanged in the results above. Re-deriving the headline figures on one machine’s terms would replace a stated, sourced assumption with an unstated generalisation from a single consumer GPU, which is the error this section exists to expose rather than repeat. What the measurement licenses is narrower than we first claimed: the harness’s timing and power instruments are mutually consistent, the modelled magnitude is right for the whole machine and too high for the accelerator alone, and any future re-parameterization now has a measured baseline to be checked against. The functional form of (2) is *not* among the things this establishes, for the reason given above.

J. H1 & H6: Transport (Supported within protocol)

The P2P expectation costs 0.48 Wh for a representative 25 MB session vs. 0.75 Wh centralized, a 36% transmission saving, small relative to inference but meaningful for media/browsing. On privacy, the WebRTC datachannel is SCTP-over-DTLS with mandatory encryption and SDP fingerprint pinning [19], [20]; a relay lacks either peer’s DTLS private key and so cannot match the pinned fingerprint; it is a blind ciphertext forwarder. Sustainability therefore does not require STUN/TURN relays to see plaintext. This does not establish application-layer privacy between paired endpoints: the authenticated peer remains a trusted endpoint, and residual relay exposure is connection metadata intrinsic to any relay.

K. What on-device adaptation changes, and what it does not

Every result above rests on the per-class capability ratios ρ_k of §IV, which this paper treats as fixed properties of parameter count. A companion study [1] tests that simplification arithmetically and now supplies a public synthetic benchmark for the narrower question of held-out task accuracy. Its three

model clusters, five classes, 20 evaluation tasks per class and three adapter seeds are useful evidence, but they do not sample the workload behind f_s or provide an independent device replication.

The crux variable is conditionally invariant at the default bar. At this paper’s acceptance threshold $\alpha = 0.70$, the declared adaptation scenario changes f_s by +0.0 percentage points: every adaptable class already cleared the bar without it, so no positive uplift can recruit further workload. That invariance is structural rather than empirical - it holds across the whole uplift range the companion study tests - and it is the finding that matters here, because it means **the headline savings of \$V are unchanged by adaptation**. What improves instead is measured correctness on the companion’s synthetic held-out tasks: 86.7% to 97.8%, or +11.1 pp. That result is not a measurement of the workload the local model was already serving. The workload-weighted 88.4% to 93.9% figure remains an exploratory scenario, and the companion study is explicit about the distinction.

Hard reasoning remains out of reach. Adaptation is modelled as reaching only classes inside the adapter’s training distribution. A small corpus contains no competition mathematics to distil, so the ceiling on the hard-reasoning class is the base model’s depth and a low-rank update does not raise it. H3’s negative finding survives the companion study intact, and the routing rule of §III is unaffected.

The device class that matters for adaptation is not the one modelled here. This paper models a discrete consumer GPU. A 27B-class adaptation job is driven to rank 32 at sequence 1024 before it fits in 24 GB, while a unified-memory APU with a quarter of the bandwidth but five times the addressable memory runs the same job at rank 128 and sequence 8192. Addressable memory, not throughput, is the binding constraint on what a household can teach; the companion study’s memory model reproduces a publicly documented 27B run to within -0.6% . Adaptation is charged its full energy cost there. Against an all-cloud counterfactual that cost repays in between three and 143 days of ordinary use; charged only against the routing the adapter actually changes, it repays nothing at $\alpha = 0.70$, for the reason just given - at that threshold it changes none. The companion study records that hypothesis as accounting-dependent rather than validated, and the distinction matters for any deployment intending to justify adaptation on footprint grounds rather than on quality.

VI. LIMITATIONS AND THREATS TO VALIDITY

We treat the counter-claims as first-class results.

C1: Rebound (Modelled, conditional). One rebound model, reported in two units, with the conversion now stated. Charging a take-back fraction r of the gross saving, $F_{\text{dec}}^{\text{eff}} = F_{\text{dec}} + r(F_{\text{cloud}} - F_{\text{dec}})$, is identical to inflating decentralized volume by $(1 + g)F_{\text{dec}}$ under

$$g = r \frac{F_{\text{cloud}} - F_{\text{dec}}}{F_{\text{dec}}}. \quad (12)$$

So the literature take-back of 15–25% [24] *is* induced extra usage of 32–53% (central 20% $\equiv$ 42%), and parity — where

the whole gross saving is given back, $r = 1$ — *is* induced usage of 212%. Both figures are reported in both units wherever they appear. Under the assumed 15–25% take-back range, rebound erodes but does not erase the modelled savings; real-world deployment rebound is not measured here and remains unresolved.

C2: Consumer-GPU inefficiency (Mechanism supported; magnitude not established here). We can now anchor this counter-claim in our own measurement rather than a third party’s. On the RTX 5070, an 8B request costs 0.65 mWh per output token on the GPU board rail, rising to 0.31 Wh for the same 300-token request once a 100 W host allowance is charged (Table VII). The mechanism behind the counter-claim is visible in our own roofline data: single-stream decode runs at 88–93% of the memory ceiling (Fig. 10), which is another way of saying that every token pays for one full pass over the weights. Batched serving amortizes that same pass across concurrent requests, so a data-centre deployment’s per-token energy falls with batch size in a way single-stream execution structurally cannot. Our AMD timing-only batch matrix provides a within-target check of this distinction: aggregate throughput at batch sizes 1, 2, 4 and 8 was compared with paired repetition IDs, but the host had no usable power sensor. A separate 10-repetition matrix for Qwen3.8 27B on the same target measured median aggregate throughput of 18.1, 18.7, 17.8 and 17.0 tok/s at those batch sizes: a small batch-2 gain followed by lower throughput at batch 4 and 8. This is still a single-target timing result, not an energy measurement or a production-serving law. *Edge therefore does not win on per-token efficiency*, and running a large model at the edge is a net loss. We decline to quote an energy ratio: the batch trace has no power measurement, and the per-token figures circulating for a data-centre deployment are not comparable to a board-rail measurement without a stated accounting boundary. The edge advantage claimed in this paper is right-sizing, water elimination, overhead removal and marginal energy — none of which is a per-token efficiency claim.

C3: Embodied carbon (Supported, conditional). On an already-owned, well-utilized device, amortized embodied carbon is $\sim 0.007 \text{ gCO}_2\text{e/query}$, negligible against $\sim 0.14 \text{ gCO}_2\text{e}$ operational; for a device bought *solely* for inference and lightly used (5×10^4 queries) it is $\sim 7 \text{ gCO}_2\text{e/query}$ and dominates. The thesis requires riding existing hardware, not provisioning new devices.

Router placement (quantified). The decision rule is not free. Every query must be classified before it can be routed, so the classifier’s own footprint is charged to every query, including the f_s majority that never reaches a data centre. Placement therefore governs whether the rule pays for itself. An on-device classifier costs 0.008 Wh, under 3% of the edge query it routes, and retains 99.3% of the modelled carbon savings. A geographically separate router costs 0.103 Wh, some 35% of an edge query, reducing savings against the GPT-4o long baseline from 54.4% to 49.7% and retaining 91.5%. In both cases the network term is negligible; the cost is the remote classifier’s own inference pass, paid on every query. The privacy

consequence is sharper than the energy one: a remote classifier is an endpoint rather than a relay, and it reads the query by construction, forfeiting the on-device confidentiality that H6 otherwise preserves. We therefore treat classifier co-location with the edge node as a constraint of the architecture rather than a deployment choice.

Other threats. Capability ratios are task-class estimates from public benchmarks; we expose α and the ratios in the sensitivity analysis and compare f_s with NVIDIA’s case-study replaceability estimates [8]. Cloud per-query energy depends on prompt length/batch; we use reported short/long figures. Network intensity spans orders of magnitude [29]; we use a conservative central value with wide bounds. We model one edge GPU class; lower-power NPUs would shift C2 favorably [17]. Finally, the current priced frontier models have no per-query energy measurement in the source set used for this release; that is a search result, not proof of absence. Their serving energy may fall anywhere between our frontier and efficient-cloud baselines. The decision rule spans that interval by construction, since it selects on the displaced baseline rather than assuming either extreme, but a measurement of current models remains the most valuable single addition to this analysis.

Scope of our own measurement (§V-H). Two machines: an AMD unified-memory APU and an NVIDIA discrete GPU. The primary energy records are single-stream; the AMD target also has a concurrent timing matrix at batch sizes 1, 2, 4 and 8, with no usable power sensor. While NVIDIA telemetry provides direct board-level power sampling and bracketed request integration, board power is not whole-system wall power: host CPU/DRAM draw and idle keep-awake power are unmeasured by GPU telemetry alone; supply conversion is now charged at an assumed $\eta = 0.90$ rather than omitted, which is an assumption and not a wall-meter reading. The board counter’s own refresh rate limits every energy figure to two significant figures (§V-H). The timing matrix is not a production-serving or energy-efficiency result: it does not measure wall power, sustained thermal behavior, queueing, or a matched cloud batch. The bandwidth ladder rests on peak figures taken from hardware specifications rather than measured transfer ceilings, and on weight-file size as a proxy for bytes streamed per token. The size of that second bias is no longer asserted to be small: it is measured at *at least* 5.7 percentage points, because a dense build lands above a ceiling it cannot exceed, and it is build-dependent rather than constant. It stays well inside the 30 percentage points separating the two decode classes, which is the only comparison the ladder is now used to support — and it is emphatically not small enough to support reading utilisation to the percentage point.

VII. DISCUSSION

The results compose into an actionable rule. Decentralized edge inference is favorable when *all* hold: (i) the task is SLM-sufficient ($\rho_k \geq \alpha$); (ii) the displaced alternative is a *frontier* model, not a hyper-efficient small cloud service; (iii) the edge device is already owned and reasonably utilized; and (iv) the

local grid is at or below the carbon crossover. When they hold, per-query carbon and water fall by roughly half to two-thirds, cost falls $\sim 150\times$, and relay plaintext access is avoided. When they do not, including hard reasoning, an already-efficient cloud baseline, a dedicated idle device, or a coal-heavy grid, route to the cloud. A heterogeneous, local-first system with capability-aware fallback [8] captures the upside while avoiding the failure modes; this is the design our reference architecture and agent implement. Folding the router into the server runtime, so that escalation follows an operator-set policy, is the natural next step; the placement result above fixes its binding constraint, that the classifier stays on the device.

Condition (i) is the one the companion study [1] reopens. Treating ρ_k as fixed makes the rule a static classification of tasks into two bins. If adaptation can move ρ_k on the classes inside a user’s own distribution, the rule becomes a function of who is asking: the same query may be SLM-sufficient for a person whose adapter has seen that kind of work and insufficient for one whose has not. That is a materially harder routing problem than the one solved here, and we flag it as open rather than claim it. The practical consequence for a deployment is narrower and immediate: because the inherited model raises no coverage at ordinary thresholds, while the synthetic benchmark shows a task-accuracy gain, a system should adapt to improve the answers it already gives locally and should not treat an adapter as a reason to widen what it attempts.

VIII. CONCLUSION

Decentralizing AI inference to already-owned edge hardware over relay-confidential P2P transport can offset a meaningful share of AI’s growing carbon, water, cost, and data-centre footprint, but only within a precisely bounded regime. The dominant lever is right-sizing (up to $65\times$), amplified by eliminating evaporative cooling and PUE/idle overhead; the binding constraints are capability sufficiency ($f_s \approx 82\%$, compared with NVIDIA’s reported 40–70%), local grid intensity, hardware reuse, and rebound. Against typical and heavy frontier usage the system saves 54–64% carbon in every declared model draw and costs $\sim 150\times$ less per query; against a best-in-class efficient cloud model it does not. A companion study [1] derives a conditional coverage result and measures a +11.1 pp gain on its synthetic held-out task benchmark. That accuracy result is limited to three model clusters on one physical host and does not identify the real-workload effect, so these bounds remain conditional. A Pass-4 measurement campaign turns the device-class argument into a within-machine test carrying its own falsification control; the premise survives, and board power on a discrete accelerator is directly measured and bracketed rather than assumed, while host-level system boundaries remain explicitly parameterized. The models, simulation harnesses and trace records behind these results are retained in full as a reusable, falsifiable basis for sustainable-AI claims, and are released openly.

The experimental benchmarking suite, simulation pipelines, parametric models, telemetry sampling harnesses, figure generators, and a runnable reference routing agent exist as a single deterministic pipeline, together with every raw per-run record and power trace behind the measurements reported here [37]. All of it is released openly under an MIT licence. All analytical models, Monte-Carlo sweeps, and empirical figures are recomputed deterministically from documented parameters, measured hardware traces, and published benchmarks, with no dependency on proprietary enterprise APIs. Hardware benchmarking operates on local execution runtimes (Ollama) under unprivileged user access, recording generation timings, prompt sweep regressions, and continuous board-level power telemetry (via the NVIDIA System Management Interface) with complete sample distributions and cache-invalidation controls. An automated verification pass extracts every quantitative token from the manuscript and confirms that each traces to a measurement record, to a declared parameter bound, or to an annotated citation. The check establishes that the prose and the underlying records agree; it does not establish that a number is used in the right sentence, and it constrains a value that coincides with many unrelated quantities only weakly. It therefore also reports, for every number, how many distinct sources could have justified it, so that the strength of each trace is visible alongside the trace itself. The routing rule of §III is implemented in the reference agent and runs live against a local SLM runtime with multi-provider cloud adapters: on a representative 10-task assistant workload it routed 80% of queries on-device, matching the modelled f_s and correctly escalating only the hard-reasoning tasks. Because generated output length varies between runs, we treat the accompanying footprint figure as a routing validation rather than a fixed measurement: the routing decisions are empirical, while per-query footprints are model-derived. A configuration audit of an operating deployment complements that simulation. Over a continuous period spanning April to July 2026, we examined the runtime logs of a single live instance of the reference platform, extracting provider-selection events alone; message payloads and user content lay outside the audit scope and were not read. Every provider-resolution event recorded in the period ($n = 97$, spanning process restarts and configuration reloads) resolved to the on-device SLM runtime; the local models configured throughout were of the 3B–8B class the analysis assumes (Ministral-3 3B and 8B, Qwen3 4B); and cloud providers remained configured and available for operator selection. As a single-instance longitudinal observation, this establishes that the local-first default is what a deployment sustains in continuous practice. It is a statement about configuration rather than per-query footprint, and one deployment rather than a sample of independent ones.

This is research release v0.3.0 dated 2026-09-14. Venue, DOI and peer-review status are intentionally unassigned in this release. Repository metadata, the citation audit, and the planned

multi-hardware and multi-model replication are released beside this manuscript.

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